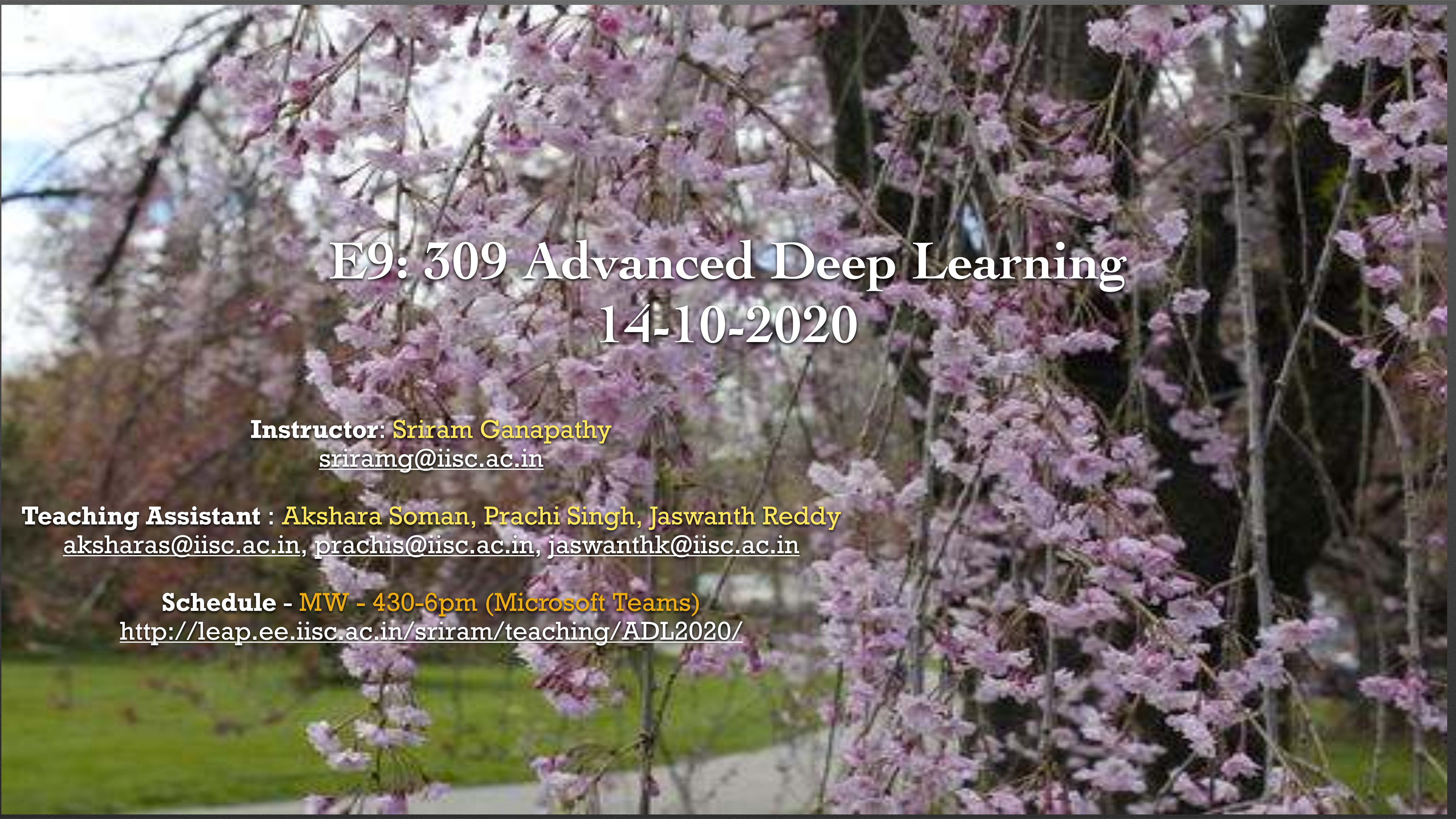




E9: 309 Advanced Deep Learning

14-10-2020

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Schedule - MW - 430-6pm (Microsoft Teams)
<http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/>

Housekeeping

✳ Filling the google form in the webpage

➡ Contents will be made available to the folks in the creditors mailing lists.

✓ Announcements regarding evaluations and projects will be shared only with creditors as well as video links.

★ Teams channel interaction and TA session for creditors only.

✳ Online registration portal from academics.iisc.ac.in

✓ Your research/faculty advisor may need to approve also before the deadline (Oct. 20th?)



Recap of previous class



Some notations

✳ $\mathbf{x} \in \mathcal{R}^D$ - input data.

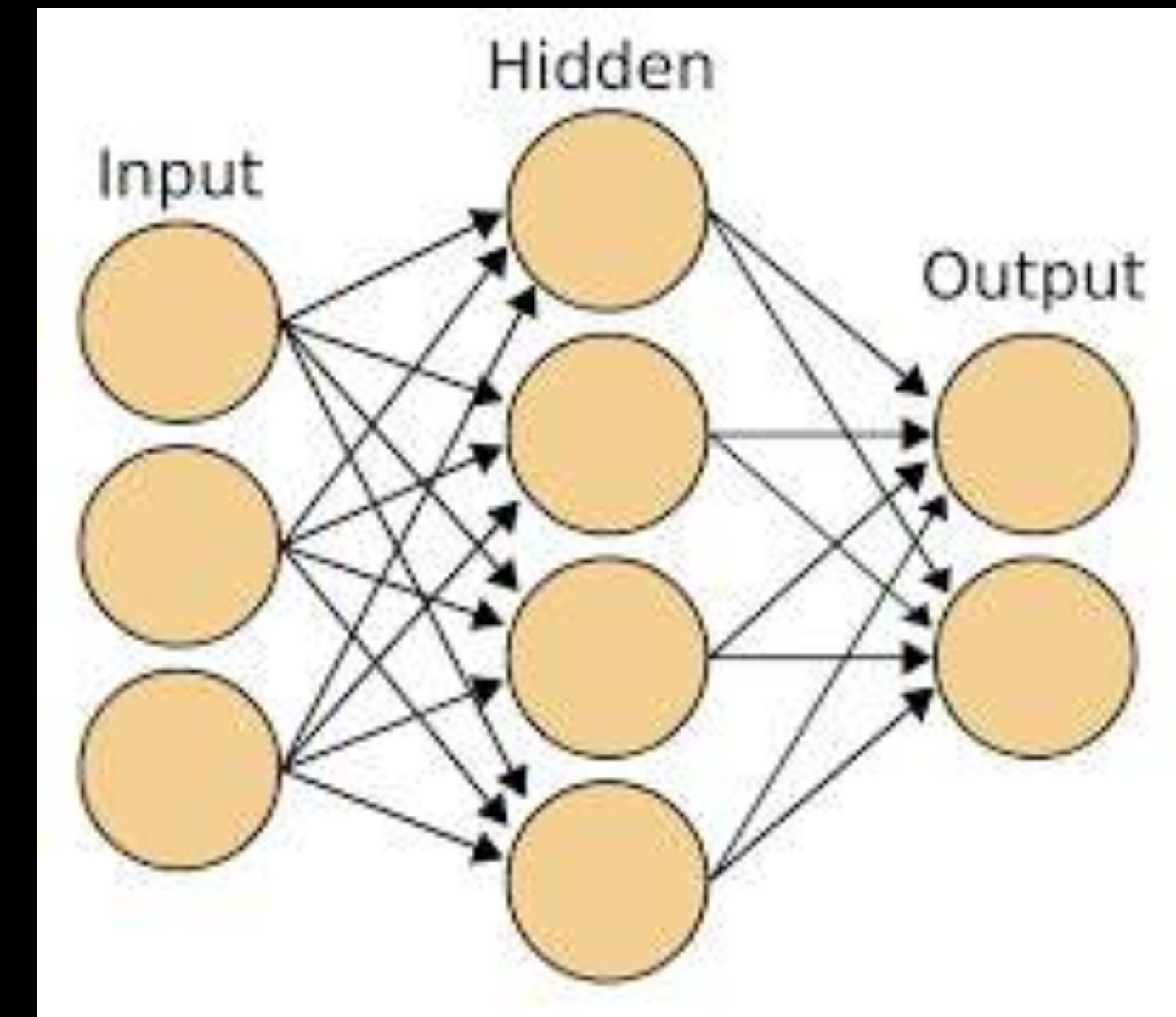
✳ $\mathbf{y} \in \mathcal{R}^C$ - neural network targets.

✳ $\hat{\mathbf{y}} \in \mathcal{B}^C$ - model outputs.

✳ $\mathbf{e}, \mathbf{h} \in \mathcal{R}^d$ - hidden model representations or embeddings.

✳ Θ - collection of learnable parameters in the model.

✳ $E(\mathbf{y}, \hat{\mathbf{y}})$ - error function used in the model training.



Some notations

✳ $\{\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{y}_1, \dots, \mathbf{y}_N\}$ - labeled training data

✳ $q = \{1 \dots Q\}$ - iteration index.

✳ $t = \{1 \dots T\}$ - discrete time index.

✳ $l = \{1 \dots L\}$ - layer index

✳ η - learning rate (hyper-parameter)

✳ N_b - mini-batch size and B is the number of mini-batches.



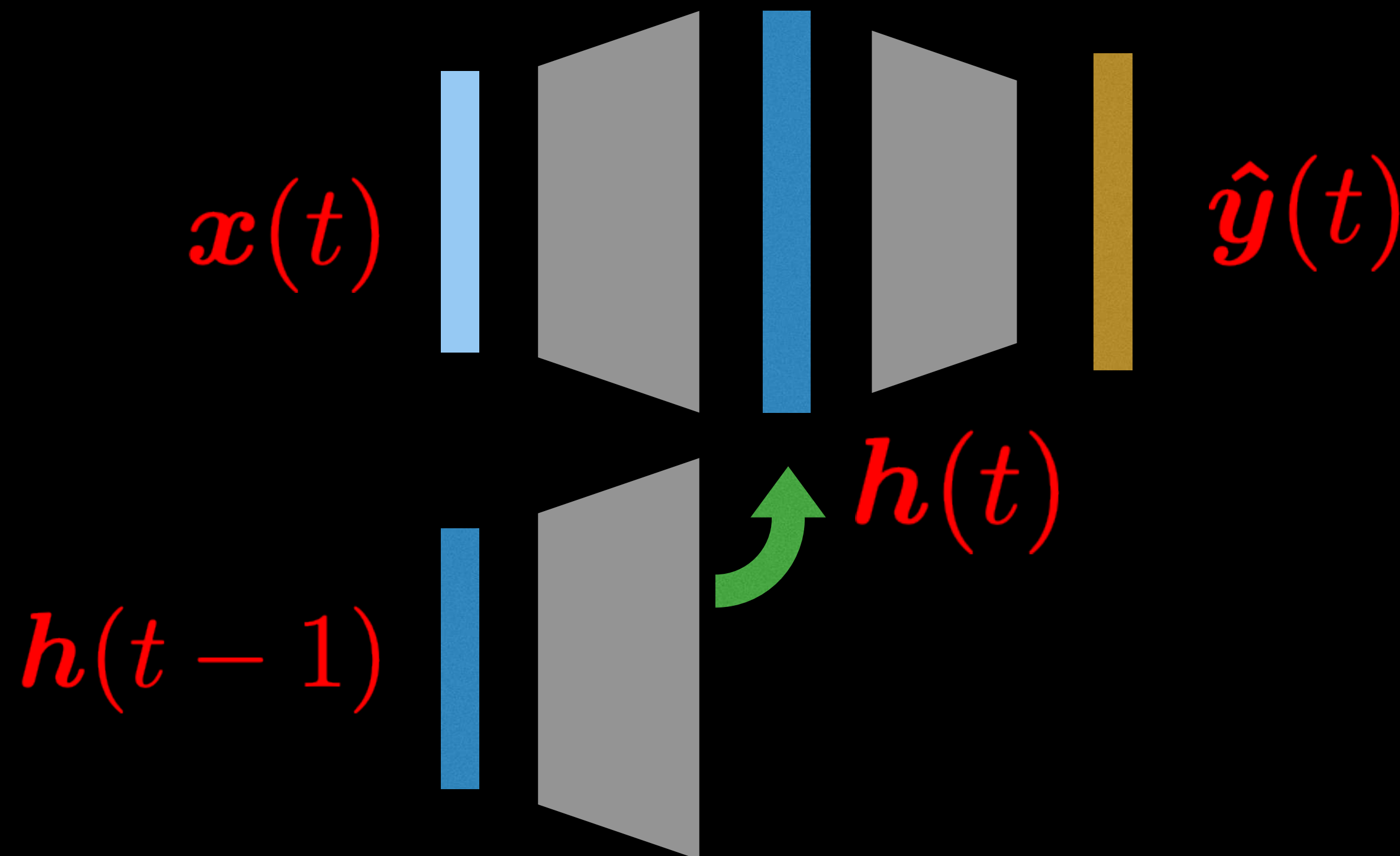
Module - I Visual and Time Series Modeling



First order recurrence - hidden layer

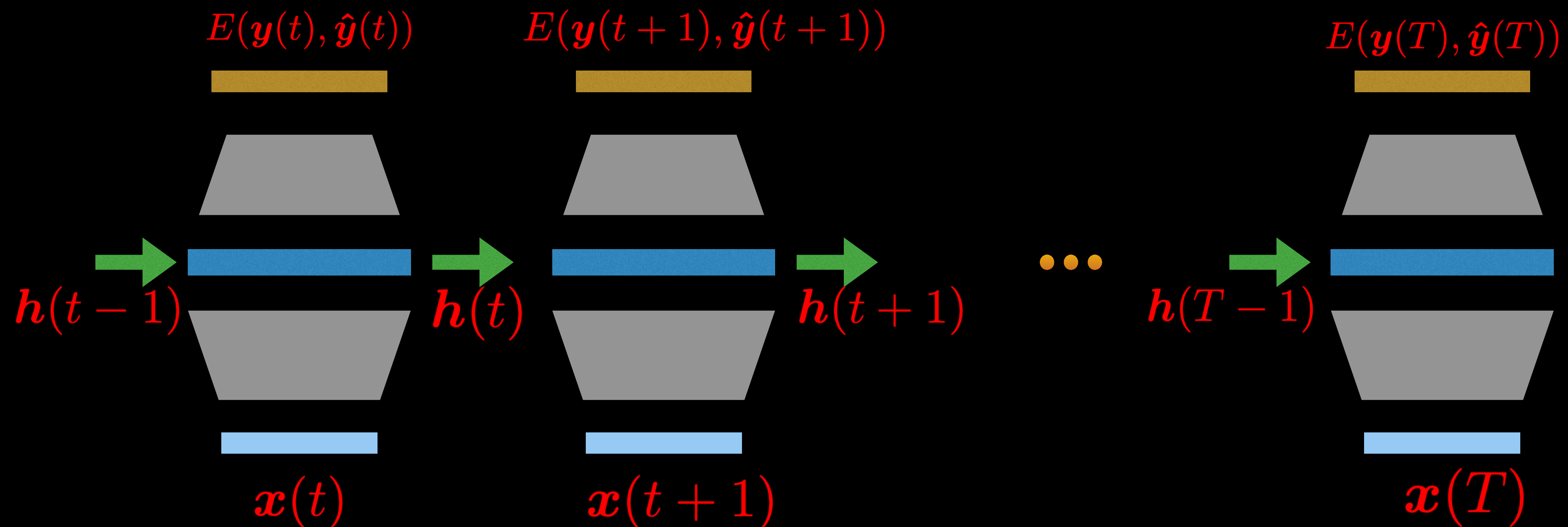
- * Making the hidden layer a function of the previous outputs from the hidden layer along with the input

$$h(t) = f(h(t-1), x(t))$$



Error backpropagation

✳ Error functions are computed at every time-instant



✳ Total error
$$E = \sum_t E(y(t), \hat{y}(t))$$

Error back propagation

Forward propagation

$$\begin{aligned} \mathbf{a}^1(t) &= \mathbf{W}^1 \mathbf{x}(t) + \mathbf{U}^1 \mathbf{h}^1(t-1) + \mathbf{b}^1 \\ \mathbf{h}^1(t) &= \tanh(\mathbf{a}^1(t)) \\ \mathbf{a}^2(t) &= \mathbf{W}^2 \mathbf{h}^1(t) + \mathbf{b}^2 \\ \hat{\mathbf{y}}(t) &= S(\mathbf{a}^2(t)) \end{aligned}$$

✓ Output activations

$$\frac{\partial E}{\partial \mathbf{a}^2(t)} = \hat{\mathbf{y}}(t) - \mathbf{y}(t) \text{ for } t = 1 \dots T$$

✓ Hidden activations at last instant T ↖

$$\frac{\partial E}{\partial \mathbf{h}^1(T)} = (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(T)}$$

✓ Hidden activations for previous instances $t = T-1, \dots, 1$

$$\frac{\partial E}{\partial \mathbf{h}^1(t)} = (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(t)} + \frac{\partial \mathbf{h}^1(t+1)}{\partial \mathbf{h}^1(t)} \frac{\partial E}{\partial \mathbf{h}^1(t+1)}$$

Softmax

cross entropy



Error back propagation

- ✓ Hidden activations for previous instances $t = T-1, \dots, 1$

$$\begin{aligned} \mathbf{a}^1(t) &= \mathbf{W}^1 \mathbf{x}(t) + \mathbf{U}^1 \mathbf{h}^1(t-1) + b^1 \\ \mathbf{h}^1(t) &= \tanh(\mathbf{a}^1(t)) \\ \mathbf{a}^2(t) &= \mathbf{W}^2 \mathbf{h}^1(t) + b^2 \\ \hat{\mathbf{y}}(t) &= S(\mathbf{a}^2(t)) \end{aligned}$$

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{h}^1(t)} &= (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(t)} + \frac{\partial \mathbf{h}^1(t+1)}{\partial \mathbf{h}^1(t)} \frac{\partial E}{\partial \mathbf{h}^1(t+1)} \\ \frac{\partial E}{\partial \mathbf{h}^1(t)} &= (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(t)} + \left((\mathbf{U}^1)^T \text{diag}(1 - (\mathbf{h}^1(t+1))^2) \right) \frac{\partial E}{\partial \mathbf{h}^1(t+1)} \end{aligned}$$

- ✓ Here, the term $\text{diag}(1 - \mathbf{h}(t+1)^2)$ comes from the derivative of $\tanh$

- ✓ and the notation $\cdot^2$ denotes element wise operation of squaring.



Error back propagation

- ✓ The derivatives of the output weights.

$$\frac{\partial E}{\partial W^2} = \sum_t \frac{\partial E}{\partial a^2(t)} (h^1(t))^T$$
$$\frac{\partial E}{\partial b^2} = \sum_t \frac{\partial E}{\partial a^2(t)}$$

$$\begin{aligned} a^1(t) &= W^1 x(t) + U^1 h^1(t-1) + b^1 \\ h^1(t) &= \tanh(a^1(t)) \\ a^2(t) &= W^2 h^1(t) + b^2 \\ \hat{y}(t) &= S(a^2(t)) \end{aligned}$$

- ✓ Transferring derivatives to the first layer

$$\frac{\partial E}{\partial a^1(t)} = \text{diag}(1 - h(t).^2) \frac{\partial E}{\partial h^1(t)}$$



Error back propagation

- ✓ The derivatives of the first layer weights.

$$\frac{\partial E}{\partial \mathbf{W}^1} = \sum_t \frac{\partial E}{\partial \mathbf{a}^1(t)} (\mathbf{x}^1(t))^T$$

$$\frac{\partial E}{\partial \mathbf{U}^1} = \sum_t \frac{\partial E}{\partial \mathbf{a}^1(t)} (\mathbf{h}^1(t))^T$$

$$\frac{\partial E}{\partial \mathbf{b}^1} = \sum_t \frac{\partial E}{\partial \mathbf{a}^1(t)}$$

$$\begin{aligned} \mathbf{a}^1(t) &= \mathbf{W}^1 \mathbf{x}(t) + \mathbf{U}^1 \mathbf{h}^1(t-1) + \mathbf{b}^1 \\ \mathbf{h}^1(t) &= \tanh(\mathbf{a}^1(t)) \\ \mathbf{a}^2(t) &= \mathbf{W}^2 \mathbf{h}^1(t) + \mathbf{b}^2 \\ \hat{\mathbf{y}}(t) &= S(\mathbf{a}^2(t)) \end{aligned}$$

BPTT
Backpropagation
through time.



Error Backpropagation

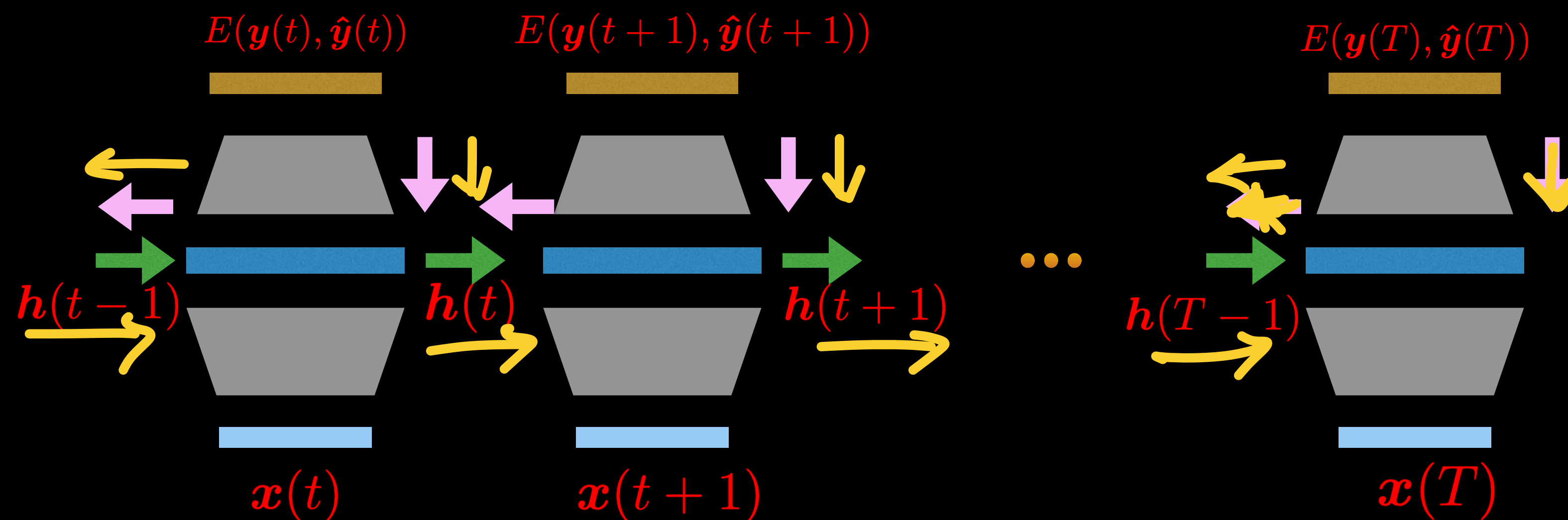
$$\begin{cases} \underline{x}'(t) = \{ \underline{x}'(t=1) \dots \underline{x}'(t=T) \} \\ \underline{x}^2(t) = \{ x^2(t) \} \\ \vdots \end{cases}$$

* Key equation in the backward direction

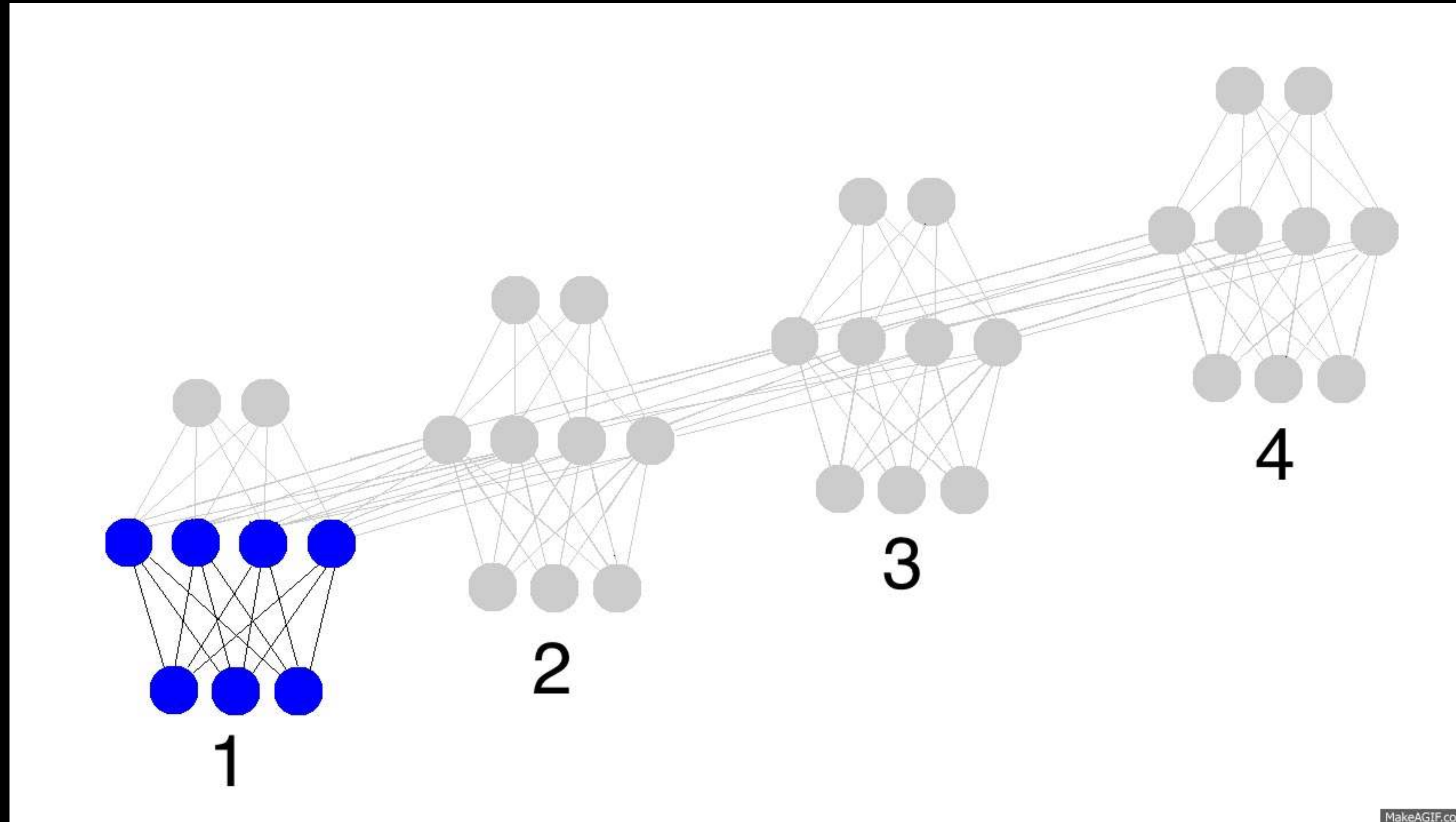
$$\frac{\partial E}{\partial \mathbf{h}^1(t)} = (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(t)} + (\mathbf{U}^1)^T \text{diag}(1 - (\mathbf{h}^1(t+1))^2) \frac{\partial E}{\partial \mathbf{h}^1(t+1)}$$

* When the model incorporates a recurrence in the forward direction.

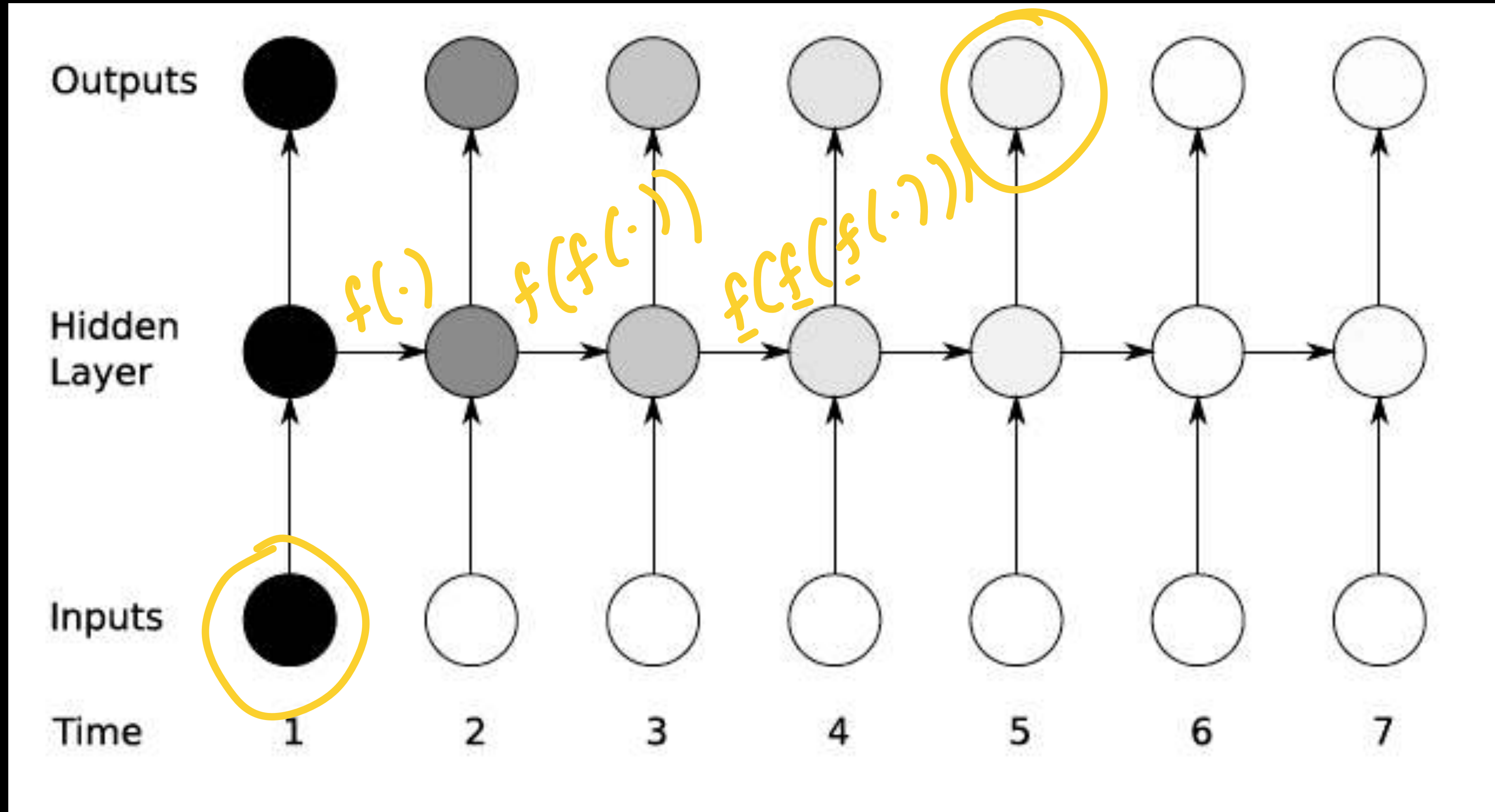
✓ Gradients incorporate a recurrence in the backward direction.



Back propagation through time



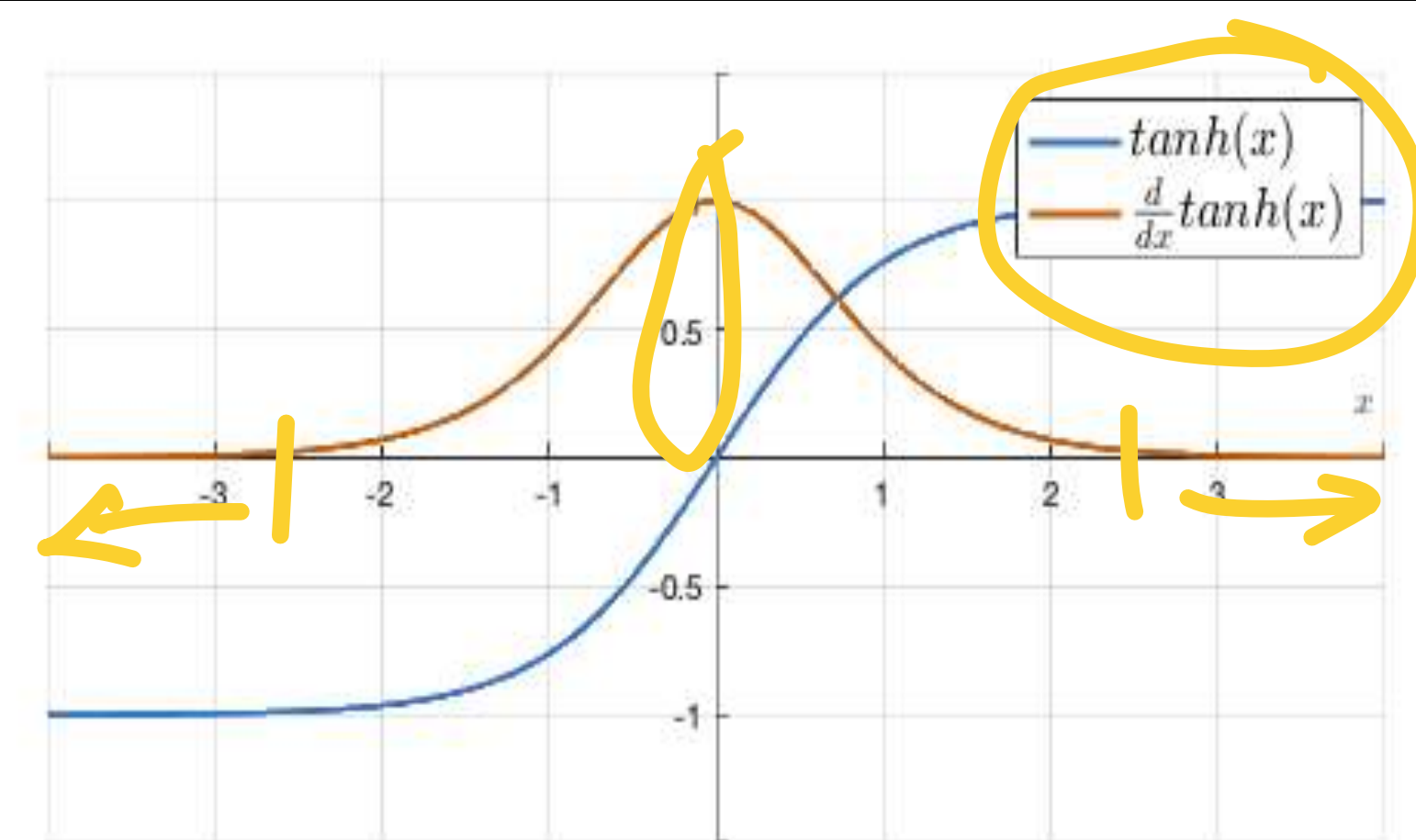
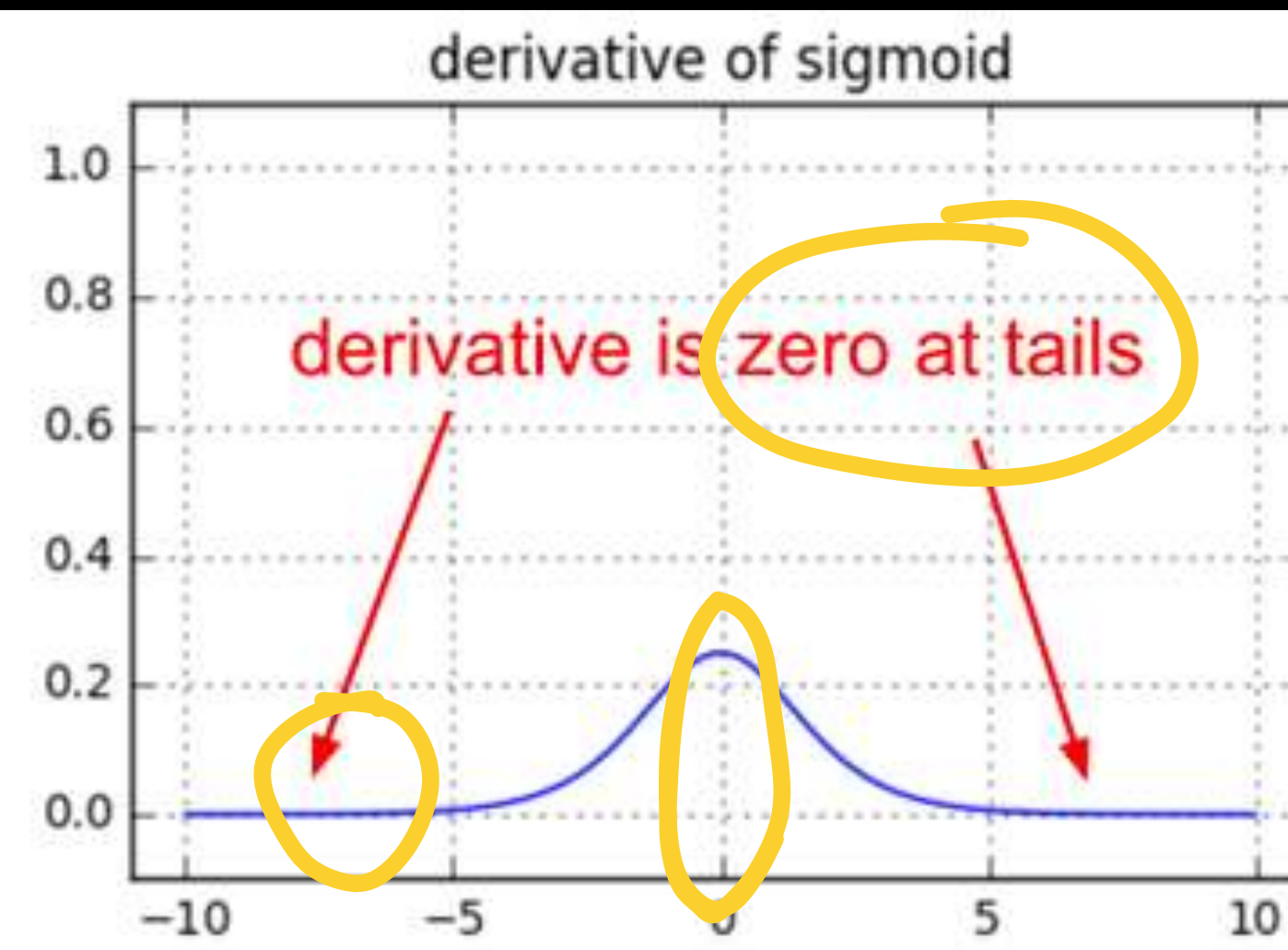
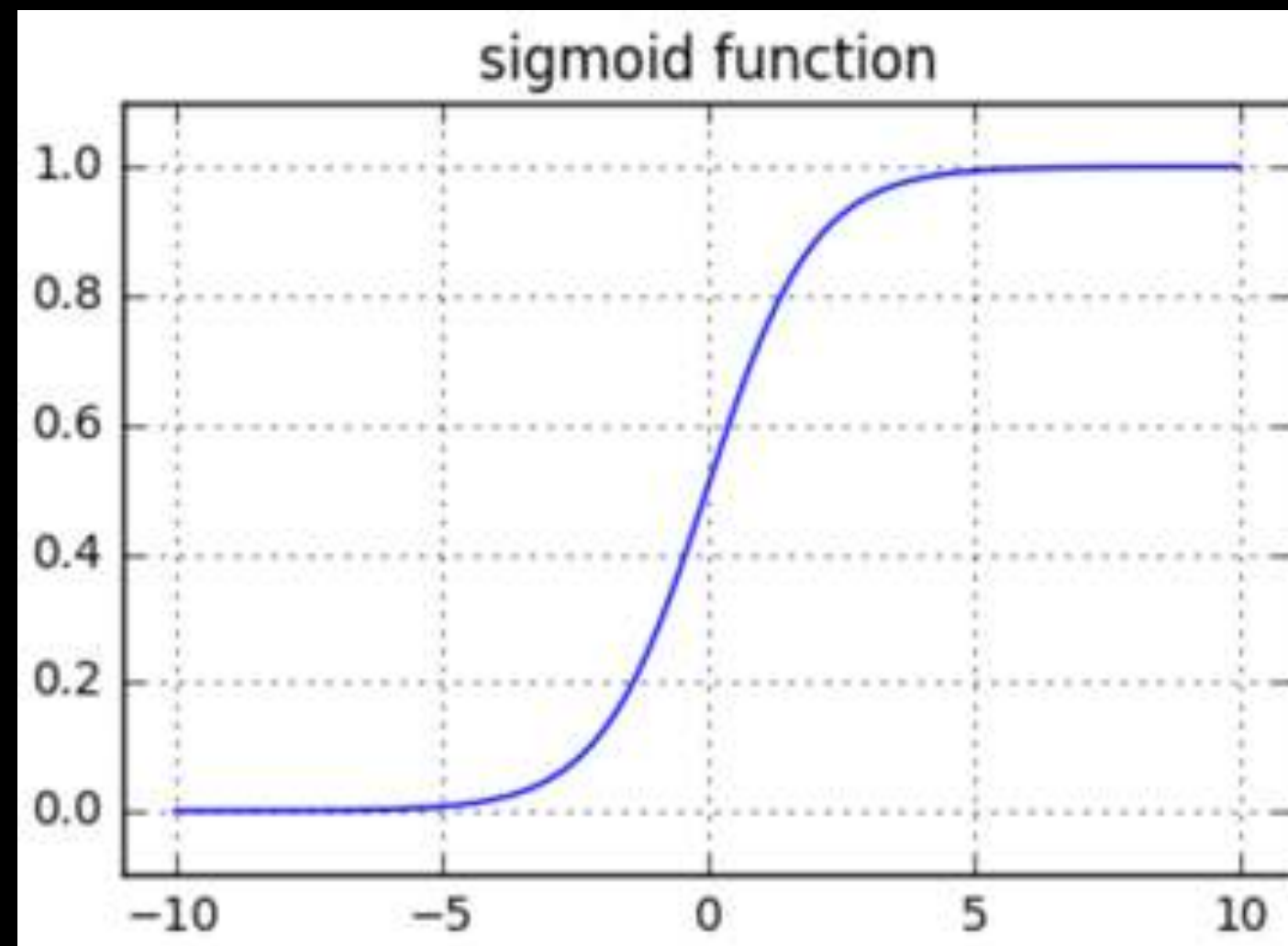
Long-term dependency issues



Long-term dependency issues

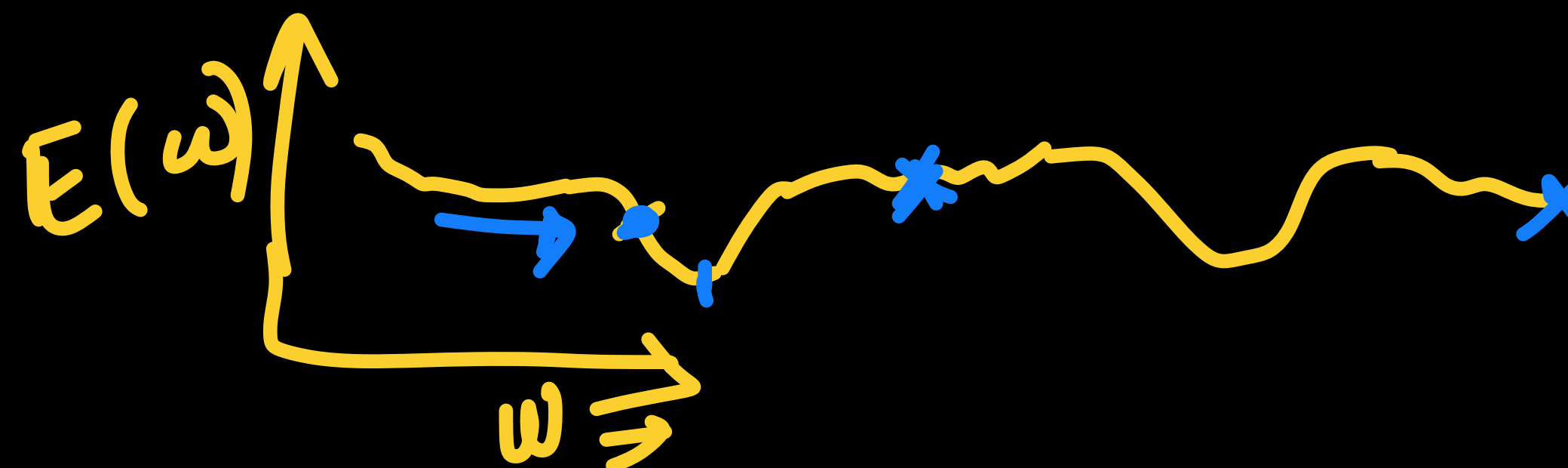
$$\frac{\partial(\hat{y}(t+10))}{\partial x(t)}$$

✓ Gradients tend to vanish or explode

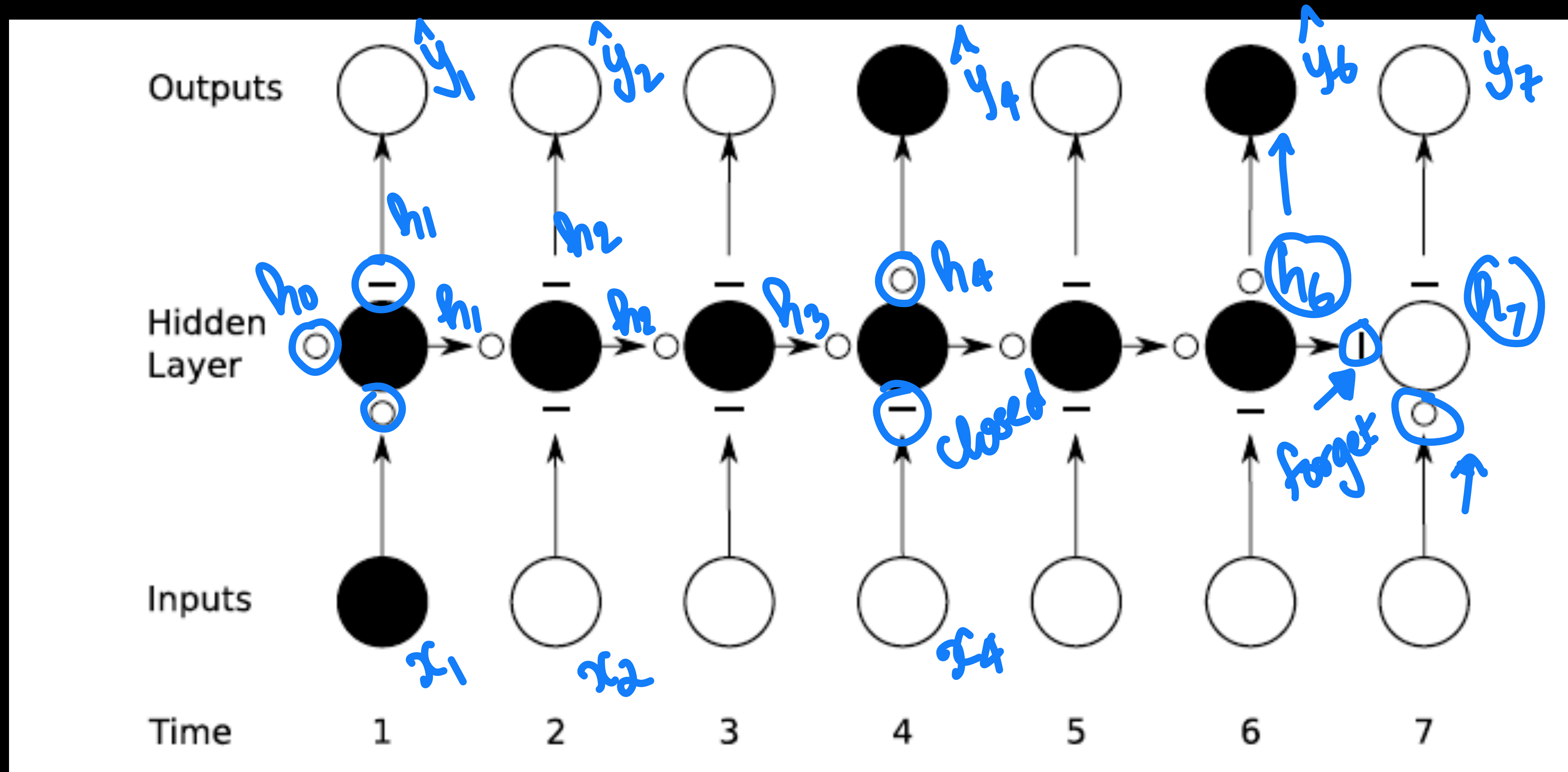


$$t = 1 \dots T \leftarrow \begin{Bmatrix} \mu \\ \sigma \end{Bmatrix}$$

✓ Initial frames may not have impact in the final predictions.



Long short term memory (LSTM) idea



Modeling questions

✳ How can we make adaptable gates with neural networks

➡ How can we make gates dependent on the data itself.

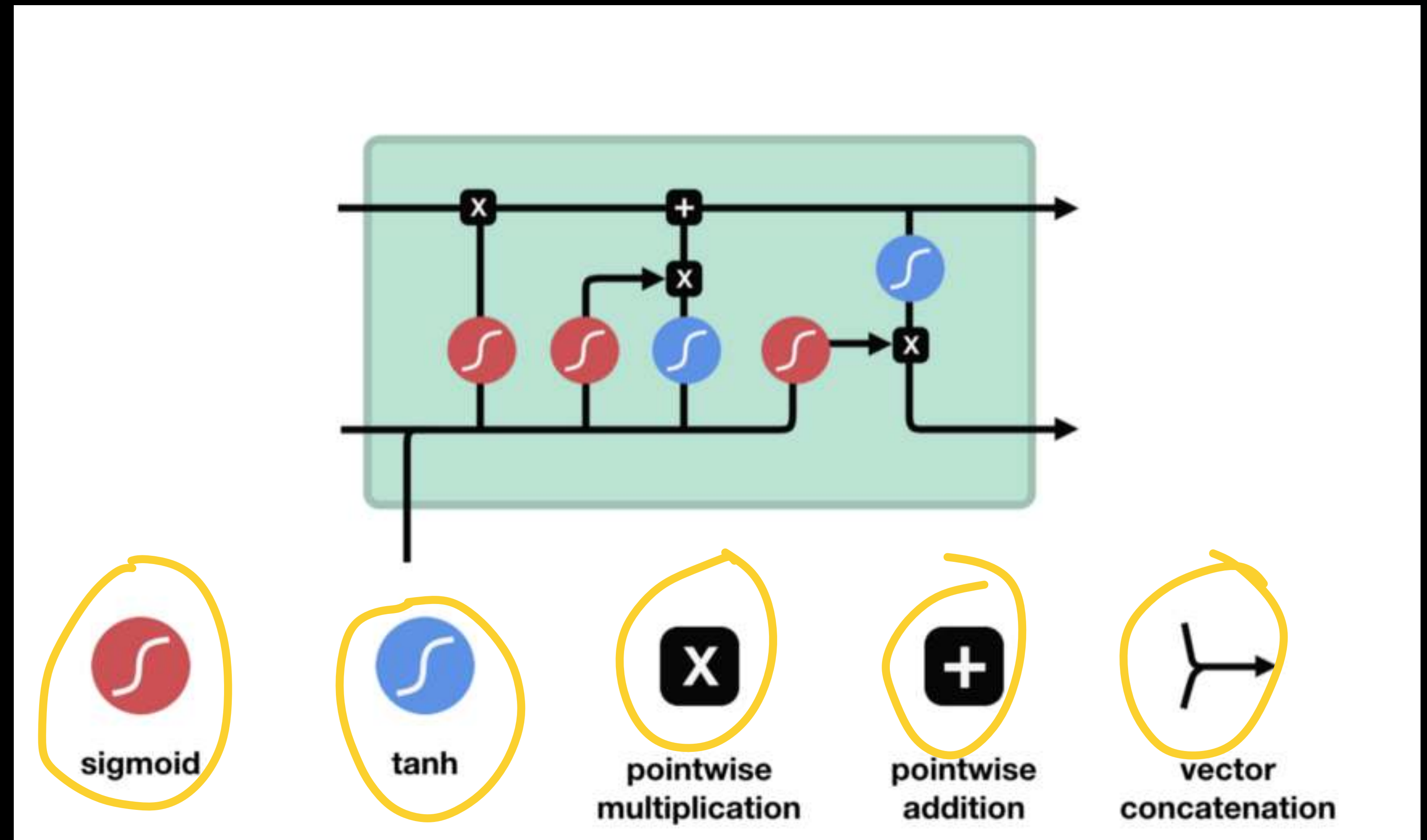
✓ Gates can be implemented as neural layers with sigmoidal outputs ?

★ Sigmoids can approximate 0-1 functions

◉ Modulate the gate output with inputs, hidden layer outputs or outputs



Long-short term memory - idea

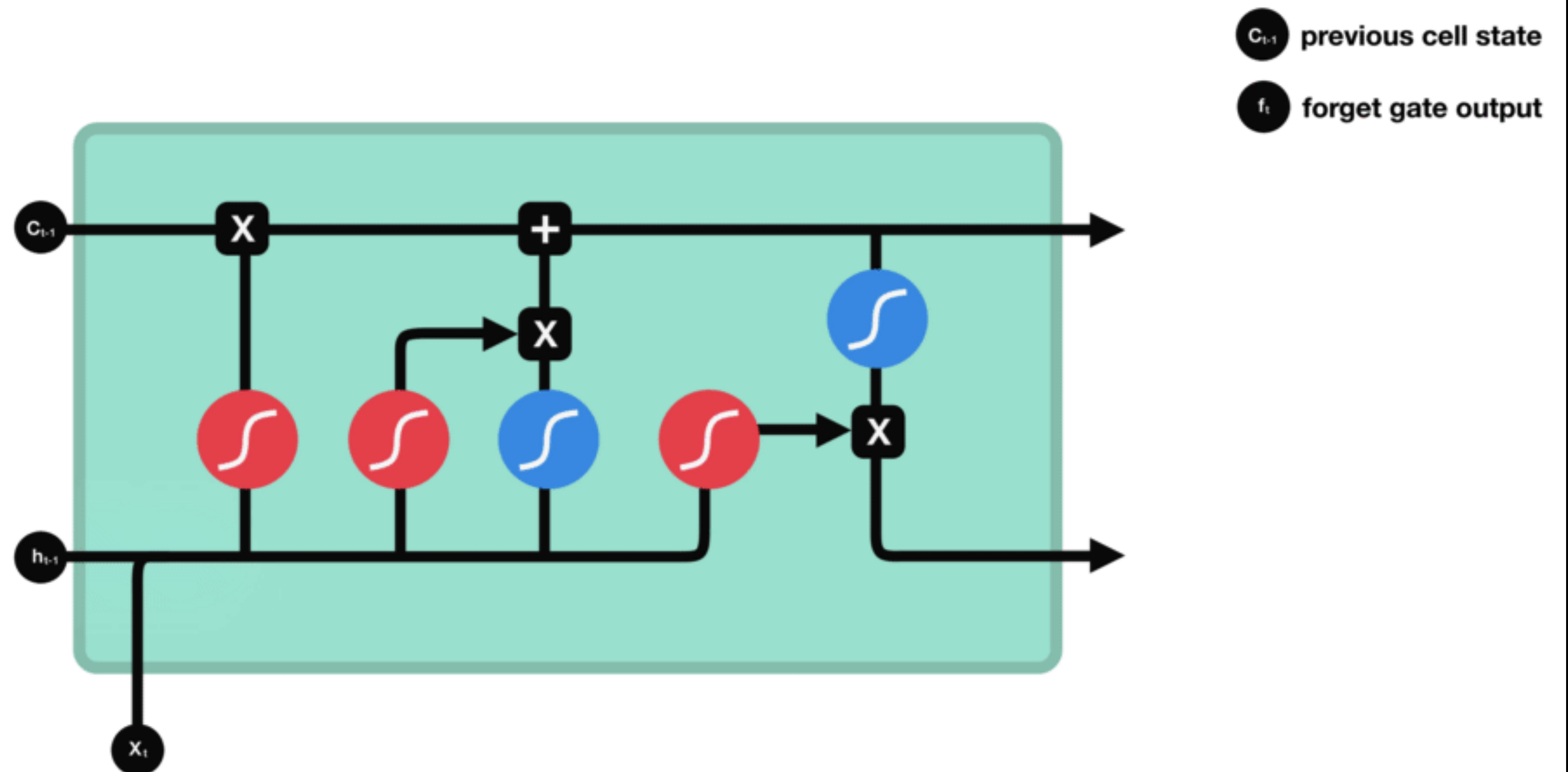


Long short term memory - idea

* Forget gate

$$a_{\phi}^t = \sum_{i=1}^I w_{i\phi} x_i^t + \sum_{h=1}^H w_{h\phi} b_h^{t-1}$$
$$b_{\phi}^t = f(a_{\phi}^t)$$

h_{t-1}

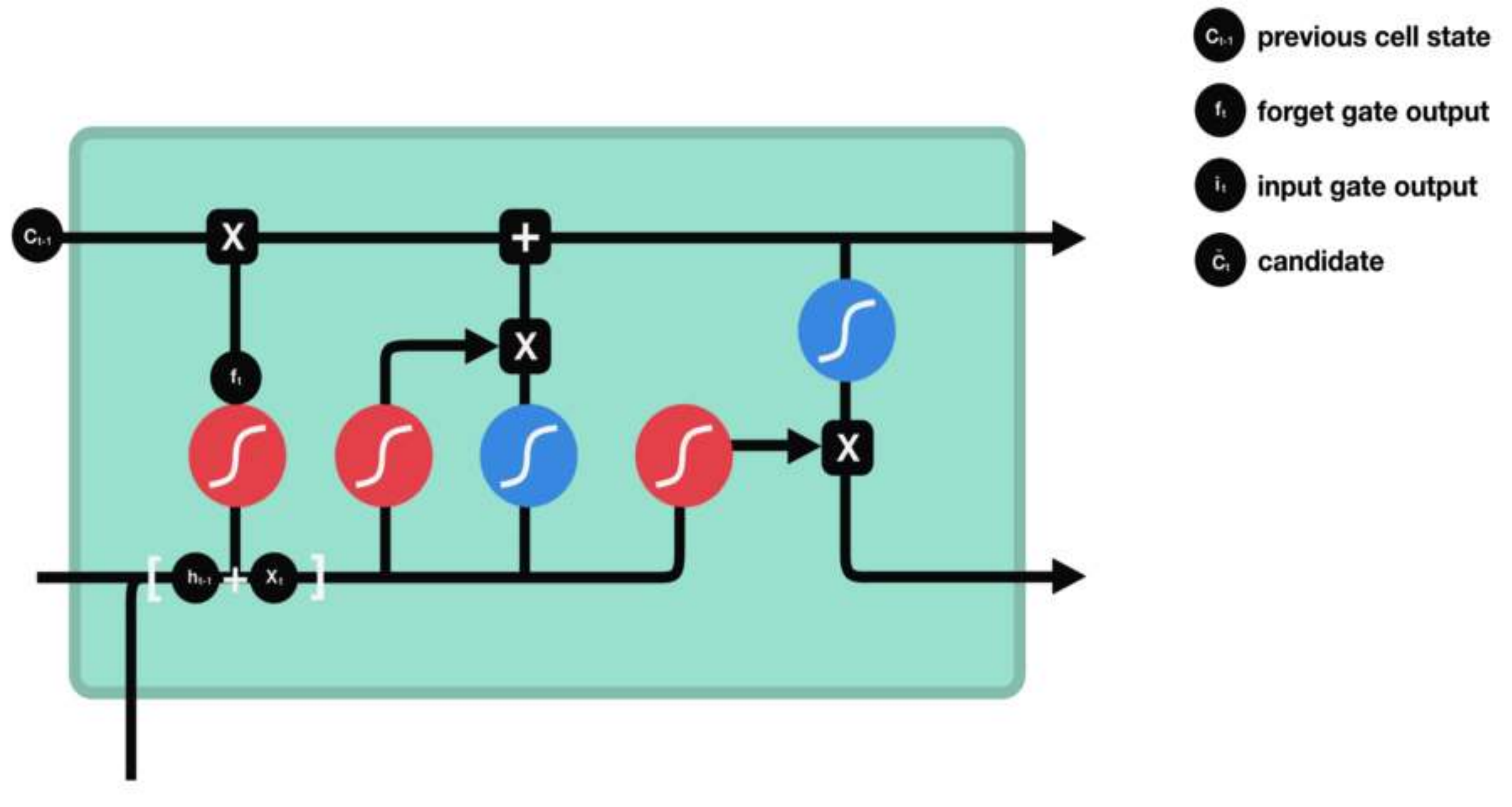


x_t

Long short term memory - idea

* Input gate

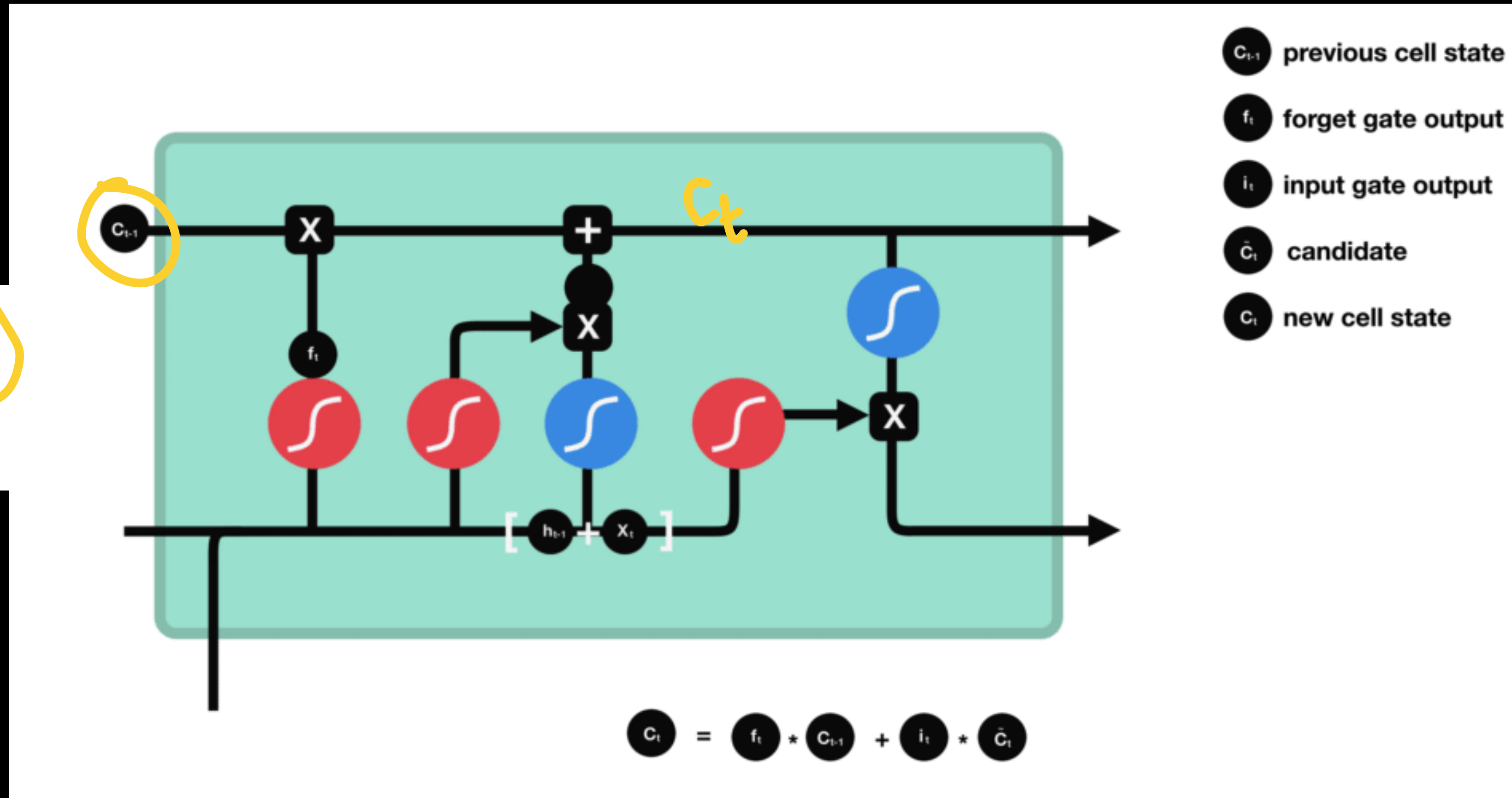
$$a_t = \sum_{i=1}^I w_{it} x_i^t + \sum_{h=1}^H w_{ht} y_h^{t-1}$$
$$b_t^t = f(a_t^t)$$



Long-short term memory - idea

✳ Cell state

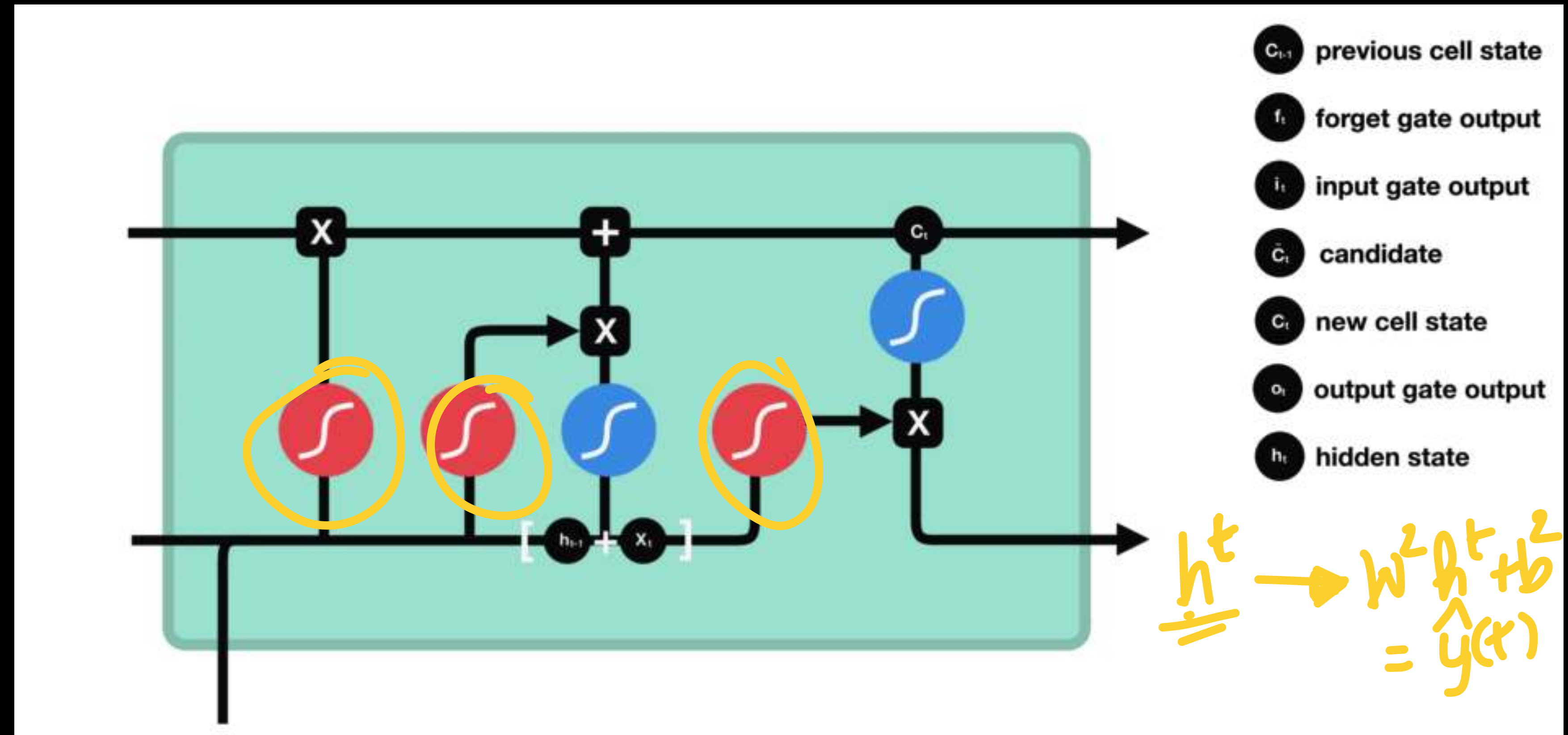
$$a_c^t = \sum_{i=1}^I w_{ic} x_i^t + \sum_{h=1}^H w_{hc} b_h^{t-1}$$
$$s_c^t = b_\phi^t s_c^{t-1} + b_i^t g(a_c^t)$$



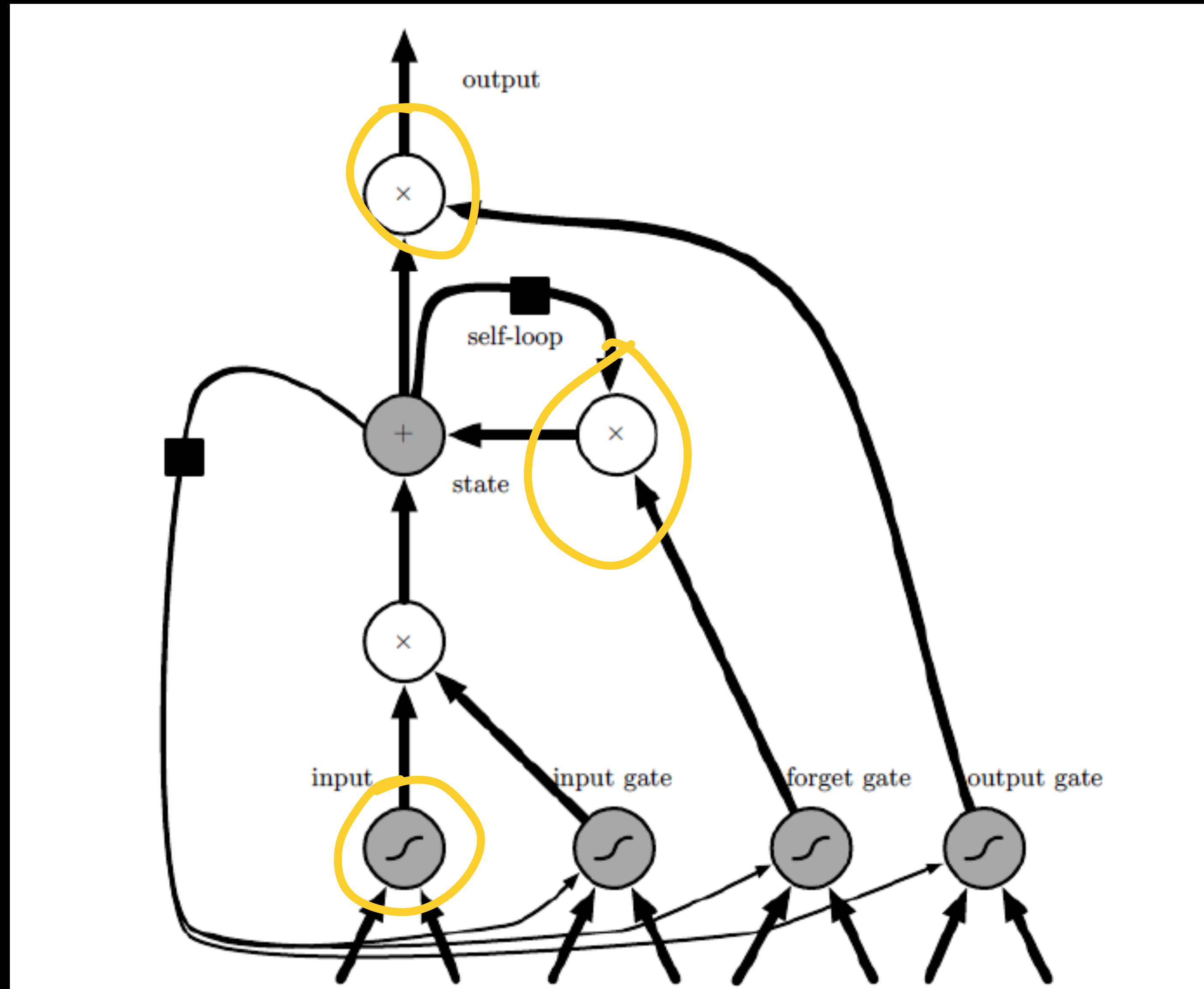
Long-short term memory - idea

* Output gate

$$a_{\omega}^t = \sum_{i=1}^I w_{i\omega} x_i^t + \sum_{h=1}^H w_{h\omega} b_h^{t-1}$$
$$b_{\omega}^t = f(a_{\omega}^t)$$



Long short-term memory - idea



Long-short term memory and GRUs

